

Subrahmanyam Doddavarapu

Senior Software Engineer

Hyderabad | +91 6305191598

subrahmanyam.embd@gmail.com

PROFESSIONAL SUMMARY

- Experienced Senior Software Engineer with 3.8 years of industry experience in network device testing, embedded testing, Sanity, Manual and Automation testing.
- Hands-on expertise in Layer 2/Layer 3 protocols including DHCP, VLAN, STP, and OSPF.
- Deployed and configured security tools such as Suricata, AIDE, USB Guard, and Clam AV for intrusion detection and endpoint protection.
- Proficient in creating comprehensive documentation, including test cases, user manuals, and process guides
- Experienced in working across diverse operating systems, including Linux and Windows.
- Proficient in MS office like word, excel, PPT and flowcharts.
- Experienced in creating and executing test cases, as well as developing test plans and test strategies.

PROJECTS

Project #1: Network Device Testing – Cisco Catalyst & ISR900 Series

Project Description: Worked on Sanity and Upgrade testing of Cisco Catalyst (Edge Routers) and ISR900 devices, executing and debugging automated test scripts using the pytest framework. Gained hands-on experience in L2/L3 protocols such as DHCP, VLAN, STP, and OSPF, along with log analysis, packet capture debugging, and collaboration with development teams for issue resolution.

Responsibilities:

- Performed Sanity and Upgrade testing on Cisco Catalyst (Edge Routers) and ISR900 series devices to validate stability, functionality, and backward compatibility across software releases.
- Executed and debugged failures in automated test scripts developed using the pytest framework, ensuring reliability and regression coverage.
- Worked on **Layer 2** and **Layer 3** protocols and features including **DHCP, VLAN, STP, and OSPF**, validating protocol behavior and network convergence in various topologies.
- Analyzed packet captures (Wireshark/tshark) and device logs for root-cause analysis of test failures.
- Collaborated with development teams to report bugs, reproduce issues, and verify fixes, contributing to continuous integration and delivery (CI/CD) pipeline improvements.
- Gained hands-on experience with network automation, Python scripting, and test framework enhancement for scaling test coverage

Operating Systems: Linux.

Project #2: Development of Smart Endoscope

Project Description: The Smart Endoscope system is used to capture images and video streaming of patients inside body on android mobile phone for doctors to treat the patient. The Smart Endoscope system Camera control circuit and android mobile with APP for user interaction.

Responsibilities:

- Conducted sanity testing on final product builds to ensure critical functionalities were stable and ready for further testing or release.
- Performed comprehensive image quality testing to ensure visual accuracy, clarity, and compliance with project standards.
- Performed functional testing to validate application features against business requirements and ensure proper system behavior.
- Developed user manuals, test case documentation, and process guides to streamline onboarding and ensure consistent project execution.

Tools: IQware, MS Office, Putty.

Operating Systems: Windows and Ubuntu.

Project #3: Olympus Gateway Box POC

Project Description: Gateway Box will provide security features to the medical device and Gateway to the cloud and Medical Device. The CV simulator application is used to simulate the Endoscope device communication with the Gateway Box. This CV simulator application is interfaced to the Gateway Box through Ethernet.

Responsibilities:

- Configured and deployed Suricata for real-time intrusion detection and network traffic analysis.
- Set up and maintained AIDE (Advanced Intrusion Detection Environment) for file integrity monitoring and alerting.
- Implemented USB Guard to control and restrict USB device access, enhancing endpoint security.
- Installed and configured Clam AV for automated malware scanning and threat detection.
- Conducted functional testing to validate the correct operation of all security tools and configurations.
- Performed regression testing to ensure new updates or changes did not affect existing functionalities.
- Created comprehensive test case documents covering various scenarios for each security tool.
- Documented configuration steps, testing procedures, and troubleshooting guidelines for future reference.
- Collaborated with the security and IT teams to align configurations with organizational policies.
- Monitored system logs and alerts to verify the effectiveness of the deployed security solutions.

SKILLS

Testing: Device Network testing, Embedded Testing, Manual Testing, Automation Testing, Device Testing, Functional Testing.

Network Protocols: DHCP, VLAN, STP, OSPF, ARP, ICMP.

Network Services: DNS, NAT, HTTP/HTTPS, FTP, SNMP.

Operating Systems: Windows and Linux.

Design and Analysis Tools: Microsoft Visio, Visual Studio Code, MS Office.

VCS: Git and GitHub, SVN.

Education Details

- Completed B. Tech (EEE) from Vishnu institute of Technology in 2021.
- Completed Diploma from Swarnandhra College of Engineering and Technology (EEE) in 2018.
- Completed SSC from Z.P.High School in 2015.

LANGUAGES

English, Telugu, Tamil.

